

Alex Oesterling

aoesterling@g.harvard.edu, alexoesterling.com

EDUCATION	Harvard University <i>Ph.D. in Computer Science</i> <i>Advised by Prof. Flavio du Pin Calmon, Prof. Hima Lakkaraju</i>	2022 – Present
	Duke University <i>B.S.E. in Electrical and Computer Engineering, Computer Science</i> <i>Summa Cum Laude</i> <i>Advised by Prof. Guillermo Sapiro, Prof. Cynthia Rudin</i>	2018 – 2022 GPA: 3.98
FELLOWSHIPS & AWARDS	National Science Foundation Graduate Research Fellowship, 2022 <i>Funding for three years of graduate studies, covering cost of tuition and stipend</i>	
	Duke George Sherrerd III Memorial Award, 2022 <i>Awarded annually by ECE faculty to the senior with the highest scholastic achievement and service to the community.</i>	
	Tau Beta Pi, 2021 <i>National engineering honor society member with high academic standing and exemplary character.</i>	
	IEEE Eta Kappa Nu, 2021 <i>IEEE honor society for electrical engineers with high academic standing and leadership potential.</i>	
	Pratt Fellowship, 2020 <i>Competitive research fellowship culminating in graduation with distinction.</i>	
	Huang Fellowship, 2019 <i>Competitive fellowship focused on science and its intersection with society.</i>	
WORK EXPERIENCE	Google DeepMind <i>Research Scientist Intern</i> Working with Google DeepMind’s image understanding team to address the modality gap problem in multi-modal models.	05/2026 - 08/2026
	Apple AI/ML Research <i>Research Scientist Intern</i> Worked with Apple’s Human Centered Machine Intelligence team to develop AI Interpretability tools to improve safety of Apple Intelligence	05/2025 - 08/2025
	Amazon Alexa <i>Software Development Engineering Intern</i> Improved ambient Alexa experience by introducing new use case for customers owning multiple Alexa devices including one device with a screen	06/2021 - 08/2021
PUBLICATIONS	“Understanding Annotator Safety Policy with Interpretability” A. Oesterling , D. Ren, Y. Assogba, D. Moritz, S.S.Y. Kim, L. Gatys, F. Hohman <i>FAccT 2026</i>	
	“Temporal Sparse Autoencoders: Leveraging the Sequential Nature of Language for Interpretability” A. Oesterling* , U. Bhalla*, C.M. Verdun, F.P. Calmon, H. Lakkaraju.	

Oral, *ICLR 2026*

*Prev. **Oral** at Workshop on Assessing World Models, ICML 2025*

“Inference Time Reward Hacking in Large Language Models”

H. Khalaf, C.M. Verdun, **A. Oesterling**, H. Lakkaraju, F.P. Calmon

Spotlight, *NeurIPS 2025*

Prev. at Workshop on Models of Human Feedback for AI Alignment, ICML 2025

“Soft Best-of- n for Model Alignment”

A. Oesterling*, C.M. Verdun*, H. Lakkaraju, F.P. Calmon

ISIT 2025

“Multi-group Proportional Representations in Text-to-Image Models.”

S. Jung, **A. Oesterling**, C.M. Verdun, S. Vithana, F.P. Calmon.

CVPR 2025

Prev. at Algorithmic Fairness through the Lens of Metrics and Evaluation Workshop, NeurIPS, 2024

“Interpreting CLIP with Sparse Linear Concept Embeddings (SpLiCE).”

A. Oesterling*, U. Bhalla*, S. Srinivas, F.P. Calmon, H. Lakkaraju.

NeurIPS, 2024

“Multi-Group Proportional Representation in Retrieval.”

A. Oesterling*, C.M. Verdun*, C.X. Long, A. Glynn, L.M. Paes, S. Vithana, M. Cardone, F.P. Calmon.

NeurIPS, 2024

“Fair Machine Unlearning: Mitigating Disparities during Data Deletion.”

A. Oesterling, J. Ma, F. P. Calmon, H. Lakkaraju.

AISTATS 2024

Prev. at Data Centric Machine Learning Workshop, ICML, 2023

“Operationalizing the Blueprint for an AI Bill of Rights: Recommendations for Practitioners, Researchers, and Policy Makers.”

A. Oesterling*, U. Bhalla*, S. Venkatasubramanian, H. Lakkaraju

Preprint

“All Roads Lead to Rome? Exploring Representational Similarities Between Latent Spaces of Generative Image Models”.

C. Badrinath, U. Bhalla, **A. Oesterling**, S. Srinivas, H. Lakkaraju.

HiLD, GRaM, SPIGM Workshops, ICML, 2024

“Multitask Learning for Citation Purpose Classification.”

A. Oesterling*, A. Ghosal*, H. Yu*, R. Xin*, Y. Baig*, L. Semenova, C. Rudin.

Second Workshop on Scholarly Document Processing (SDP), NAACL, 2021

“Distributionally Robust Group Backwards Compatibility.”

M. Bertran, N. Martinez, **A. Oesterling**, and G. Sapiro.

DistShift Workshop, NeurIPS, 2021

“Detecting Motion in a Room Using a Dynamic Metasurface Antenna.”

A. Oesterling, M. Imani, O. Mizrahi, J. Gollub, and D. Smith.

IEEE Access, 2020

“Integrated Single-cell Multiomic Analysis of HIV Latency Reversal Reveals Novel Regulators of Viral Reactivation.”

A. Manickam, J. Peterson, Y. Harigaya, D. Murdoch, D. Margolis, **A. Oesterling**, Z. Guo, C. Rudin, Y. Jiang, E. Browne.

Genomics, Proteomics & Bioinformatics, 2024

“PP 1.33 – 00167 Integrated single-cell multiomic profiling of HIV latency reversal.”

A. Manickam, J. Peterson, W. Mei, D. Murdoch, D. Margolis, **A. Oesterling**, Z. Guo, C. Rudin, Y. Jiang, E. Browne.

Journal of Virus Eradication, 2022

** denotes equal contribution*

INVITED TALKS	Oral Presentation , ICLR	4/2026
	Workshop on Assessing World Models , ICML	7/2025
	Algorithmic Alignment Lab , MIT	12/2024
TEACHING	Lecturer , <i>CS 2901/2: Seminar on Effective Research Practices and Academic Culture</i>	Harvard, Fall 2025, Spring 2026
	Teaching Fellow , <i>ES 156: Signals and Communications</i>	Harvard, Spring 2024
	Head Teaching Assistant , <i>First Year Design (EGR 101)</i>	Duke, Fall 2020-2021
	Teaching Assistant , <i>First Year Design (EGR 101)</i>	Duke, Fall 2019
SERVICE	Program Chair <i>RegML @ NeurIPS 2024</i>	
	Area Chair <i>RegML @ NeurIPS 2025</i>	
	Conference Reviewer <i>NeurIPS, ICML, ICLR, FAccT</i>	
	Top Reviewer Award <i>Complimentary Registration, NeurIPS 2025</i>	
	Journal Reviewer <i>IEEE Journal on Selected Areas in Information Theory</i>	
LEADERSHIP	Brownstone Residential Living Group <i>President, Historian</i>	01/2019 – 05/2022
	Duke Pureun KPop Dance Team <i>Social Chair and Small Group Leader</i>	09/2019 – 05/2022